HYPERLOGISTICS-STREAM: REAL-TIME HUB-TO-HUB ARRIVAL DELAY PREDICTION

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UMKC

INNOVATION
Live and historical data in **one pipeline**: today's hub conditions compared with a **typical day** for that hub.

HOW WE DIFFER
Status apps show only *now*; delay statistics only *the past*. We join both and **state where the model fails**.

PRACTICAL IMPACT
Late departures often arrive less late: **77%** of live flights made up time. Better **pickups and connections**.

Problem Statement



- A late flight leaves **travelers and shipments** waiting at the next hub. Weather and ground stops make it worse.
- **Goal:** predict Δ in **real time** and flag hubs worse than their **typical** day.
- Flights are a **public proxy** for hub-to-hub freight.

Two Data Worlds

HISTORICAL: train and test

BTS On-Time flights
2018–2024, 24 US hubs. Train 8.0M, test 1.67M.

NOAA ISD weather
Hourly wind, visibility, ceiling, temperature.

LIVE: today

FlightAware AeroAPI
Scheduled vs actual: true delay on landing.

FAA NAS status
Ground stops and delay programs per airport.

NWS weather
Live observations at each hub.

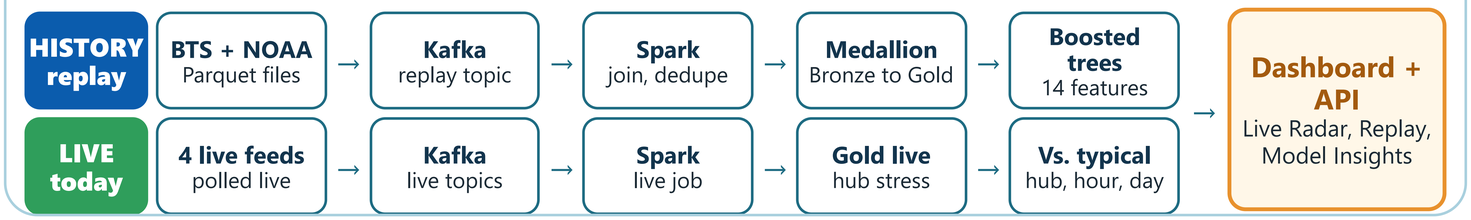
OpenSky aircraft
Aircraft near hubs (no delay minutes).

They share hub, hour, weekday and weather: the bridge for comparing today with the past. **Only history** has delay minutes for millions of flights, so the model trains there.

How Our Approach Differs

	Flight-status apps	Delay-statistics sites	HyperLogistics-Stream
Uses live data now	yes	no	yes
Uses years of history	rarely	yes	yes (8M flights)
Compares today with a typical day	no	no	yes, per hub, hour, weekday
Estimates arrival from delay and weather	airline estimate	no	yes, 14 inputs
Open, tested, states its limits	no	partly	60+ tests, limits listed

Architecture: One Pipeline, Two Data Worlds



The Models and How We Evaluate Them

1. No change
Arrival delay = departure delay. What travelers assume today.

2. Lane mean
Average change on that lane in the training years.

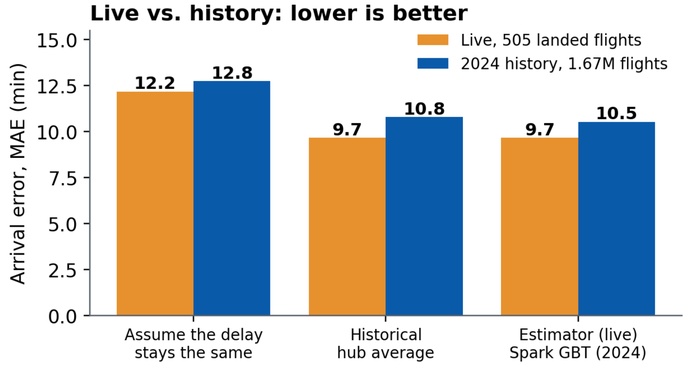
3. Boosted trees, 14 features
Gradient-boosted trees (GBT) in Spark MLlib: delay, distance, hours, weekday, lane, weather.

4. Live estimator
Rule-based stand-in for live flights: hub and hour averages. Not the GBT.

Test on a year the model never saw. Train 2018–2023 (8.0M flights), test 2024 (1.67M). **Mean absolute error (MAE)** = average arrival-time error in minutes; lower is better. All models are scored on the same flights.

12.75	10.80	10.53
no change	lane mean	GBT: 17% better

Live vs. Historical: Does the Pattern Hold Today?

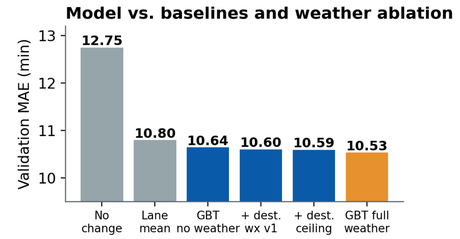


What it means for travelers

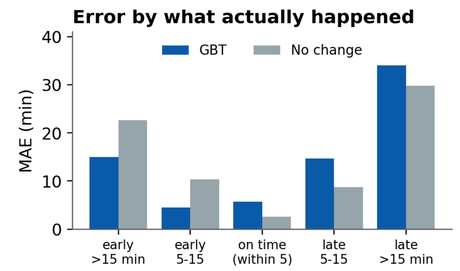
- **77% of 505 landed live flights made up time** in the air (average −8 min vs. departure delay).
- "Same delay" is off by **12.2 min**; adjusting with history, **9.7 min** (**21% closer**).
- Time **pickups and connections**; spot hubs worse than typical.

Caveat: one evening, 505 flights, rule-based estimator, not the Spark GBT. Big delays are still missed.

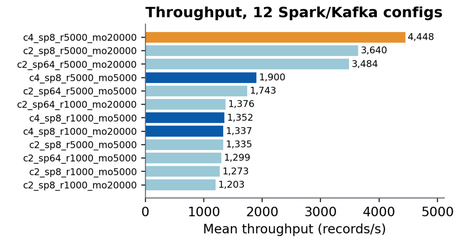
Results & Evaluation



GBT beats both baselines on 2024. Weather adds about 1%; lane and schedule matter most.



Wins on early flights, loses to "no change" on on-time and late ones. Not a severe-delay alarm.



4,448 records/s best (orange); 4 cores (dark) beat 2 by ~22%. Provisional: 4 batches per config.

Conclusion & Future Work

- **Impact:** live and historical data in one repeat-safe pipeline, 60+ tests.
- **Limits:** big delays are missed; the live check is small.
- **Next:** score live flights with the boosted-trees model.
- **Thanks** to Dr. Md Yusuf Sarwar Uddin for feedback that shaped this project.